

Pathogenicity, Development, and Reproduction of *Heterorhabditis bacteriophora* and *Steinernema carpocapsae* under Axenic *In Vivo* Conditions

Richou Han and Ralf-Udo Ehlers

Institute for Phytopathology, Department for Biotechnology and Biological Control, Christian-Albrechts-University Kiel,
Klausdorfer Strasse 28-36, 24223 Raisdorf, Germany
E-mail: ehlers@biotec.uni-kiel.de

Received December 3, 1998; accepted August 16, 1999

Galleria mellonella larvae cultured axenically were treated with axenic dauer juveniles of *Heterorhabditis bacteriophora* and *Steinernema carpocapsae*. After 3 days *S. carpocapsae* had killed all insects, with 9.4 ± 4.3 nematodes per larva. *H. bacteriophora* were unable to kill *G. mellonella*, although 13.3 ± 6.4 nematodes per *Galleria* were found in the hemocoel. Invading nematodes of both strains recovered from the dauer stage. *H. bacteriophora* developed into hermaphrodites with eggs and J1 in the uterus and in the hemolymph of the living insects. Development beyond the J1 stage was not recorded. An injection of supernatants from different *Photorhabdus luminescens* cultures killed the insects but could not provide nutrients to support a further development. Only the injection of bacterial cells supported production of dauers in the axenic insects. Axenic *S. carpocapsae* developed to adults and produced offspring. After 3 weeks an average of 5275 nematodes per larva were counted, of which 6.7% were dauer juveniles, 39.2% other juvenile stages, 11.9% males, and 42.2% females. Compared to *in vivo* reproduction in the presence of the symbiotic bacterium *Xenorhabdus nematophilus* the dauer juvenile yields were low. Even after 5 weeks the percentage of dauer juveniles did not surpass 10%. © 2000 Academic Press

Key Words: *Heterorhabditis bacteriophora*; *Steinernema carpocapsae*; symbiosis; *Photorhabdus luminescens*; *Xenorhabdus nematophilus*; *X. poinarii*; *Galleria mellonella*; axenic culture; nutritive significance.

INTRODUCTION

Entomopathogenic nematodes of the families Steinernematidae and Heterorhabditidae are associated with symbiotic bacteria of the genera *Xenorhabdus* and *Photorhabdus*, respectively. They have received considerable attention as effective and safe biological control agents for a variety of insect pests (Bedding *et al.*, 1993;

Ehlers, 1996; Ehlers and Hokkanen, 1996). The free-living and nonfeeding dauer juvenile (DJ) carries cells of its symbiont in the anterior part of the intestine, protecting the bacteria from competitive conditions in the soil environment. After entering an insect host the DJs release the cells into the sterile insect's hemolymph. Providing that the insect's humoral and cellular defense mechanisms are overcome, the insect is killed within a few days. The bacteria proliferate; the nematodes feed on the bacterial cells and metabolized host tissue and produce offspring. After two or three generations DJs leave the insect cadaver searching for new hosts.

Poinar and Thomas (1966) investigated the significance of *X. nematophilus* in the development of *S. carpocapsae* strain DD-136 in axenic *Galleria mellonella*. They discovered that the nematode was able to kill the insect even without the presence of any bacteria. Once the dauer juvenile had reached the host hemolymph, development to adult stages was observed; however, reproduction was recorded only when bacteria were added. *In vitro* combinations of *H. bacteriophora* with nonspecific *P. luminescens* resulted in bacteria-free DJs. When septic *G. mellonella* were exposed to these DJs, the insects were not killed, although DJs were able to invade the insects (Han *et al.*, 1991; Gerritsen and Smits, 1993). *In vitro* only *Steinernema* spp., but not *Heterorhabditis* spp., were successfully cultured axenically (Lunau *et al.*, 1993). However, the production of bacteria-free *H. bacteriophora* and *H. indica* dauer juveniles is possible when nematodes are cultured with nonspecific *P. luminescens* symbionts (Han and Ehlers, 1998). In order to further elucidate the nutritive dependence of entomopathogenic nematodes on their symbiotic bacteria this investigation describes the development and pathogenicity of bacteria-free *H. bacteriophora* and *S. carpocapsae* dauer juveniles in axenic *G. mellonella*.

MATERIALS AND METHODS

Axenic Insect Culture

G. mellonella were grown axenically at 35°C in glass jars closed with a triple layer of filter paper on a sterilized artificial medium containing 22% ground maize, 22% ground wheat, 11% milk powder, 5.5% yeast extract, 17.5% bee wax, 11% honey, and 11% glycerol. Still hot after sterilization the jars were shaken in order to obtain small medium pellets. Prior to inoculation the insect eggs were sterilized with 1% sodium hypochlorite for 30 min and then rinsed in sterile distilled water. The absence of xenic organisms in *Galleria* larvae was checked by streaking dissected larvae onto Standard I Agar (Merck) or by transfer of dissected larvae into YS broth (Dye, 1968) and incubation at 200 rpm at 25°C for several days.

Bacterial Cultures

Bacterial isolates were obtained from the hemolymph of last instar *G. mellonella* 48 h after infection with nematode dauer juveniles. Strain designation and origin of the nematode symbionts used for the studies are listed in Table 1. They were cultured in YS broth at 25°C and 200 rpm. The primary form of the bacteria was obtained by selecting green or blue colonies on NBTA and red colonies on MacConkey agar (Akhurst, 1980). Secondary form HNA were obtained according to Krasomil-Osterfeld (1995). Stock cultures were maintained at -80°C in 15% glycerol (v/v) in LCM (liquid culture medium) containing (in g/L) 10 tryptic soy broth (bioMerieux), 10 nutrient broth (Difco), 5 yeast extract, 5 casein peptone, 0.35 KCl, 0.21 CaCl₂, 5 NaCl (Merck), and 30 ml corn oil (Azco) (Ehlers *et al.*, 1998).

Production of Axenic Dauer Juveniles

Axenic DJs of *H. bacteriophora* H06 and *S. carpocapsae* A24 were obtained from liquid culture in LCM according to the method described by Han and Ehlers (1998). Briefly, *H. bacteriophora* and *S. carpocapsae* were grown monoxenically in LCM on bacteria isolated from *H. megidis* HNA and *S. glaseri* RS92, respectively. Molecular identification according to the method described by Ehlers and Niemann (1998) revealed that

the *P. luminescens* symbiont of the HNA strain belongs to the group *heliothidis*, containing symbionts of *H. bacteriophora* isolated from the northeastern United States. It could not be assigned to the European group *temperatus*, which harbors symbionts of other *H. megidis*. Using these combinations with nonspecific bacteria, the nematodes can reproduce but the DJs are unable to retain the bacteria in their intestine. After surface sterilization they are free of any bacteria or other microorganisms. DJs were collected by centrifugation and cloth migration through a 30-μm sieve under sterile conditions and then surface-sterilized in 0.5% streptomycin sulfate (Merck) for 6 h. The sterilization solution was removed by centrifugation, and the DJs were rinsed three times in sterile Ringer's solution. For checking the success of the surface sterilization, DJs were transferred into sterile YS broth.

Pathogenicity of Axenic Dauer Juveniles in Axenic *G. mellonella* Larvae

To test the pathogenicity of axenic nematodes, aseptic last instars of *G. mellonella* were placed on a sterile filter paper in a sterile petri dish (*d* = 6 cm). Axenic dauer juveniles (*n* = 150) in 0.1 ml sterile Ringer's solution were added to five insects. Surface-sterilized DJs of strain H06 reared with their indigenous bacterium were used as a positive control. In another control *G. mellonella* larvae received 0.1 ml sterile Ringer's solution without nematodes. The insects were incubated at 25°C. After 3 days the larvae were removed from the filter paper and incubated separately. Insect mortality was recorded every 3 days until 40 days after inoculation.

Development of Axenic Dauer Juveniles in Axenic *G. mellonella* Larvae

Three days after inoculation with nematodes and then at 3- to 5-day intervals dead and living insect larvae were randomly selected and dissected. Total number and developmental stages were determined under the microscope. Recovery (exit from DJ stage) of DJs in axenic insect larvae was determined after 3 days according to Strauch and Ehlers (1998). In order to investigate whether bacterial culture supernatant can promote the development of *H. bacteriophora* under

TABLE 1
Strain Designation and Origin of Nematode Symbiotic Bacteria *Photorhabdus luminescens*, *Xenorhabdus nematophilus*, and *X. poinarii*

Species	Strain	Nematode	Origin	Received from
<i>P. luminescens</i>	H06	<i>H. bacteriophora</i>	China	—
<i>P. luminescens</i>	HNA	<i>H. megidis</i>	Ohio, USA	Wouts
<i>P. luminescens</i>	LN2	<i>H. indica</i>	Tamil Nadu	Easwaramoorthy
<i>X. nematophilus</i>	A24	<i>S. carpocapsae</i>	Russia	Bedding
<i>X. poinarii</i>	RS92	<i>S. glaseri</i>	United States	Gaugler

axenic *in vivo* conditions, 20 μ l bacterial supernatant of H06 was injected into *G. mellonella* prior to and 5 days after nematode inoculation. The injections were done as described by Ehlers *et al.* (1997). Supernatant was obtained from a 48-h YS broth culture ($OD_{725} = 3.0$) which was centrifuged at 14,000g for 10 min at 4°C and then filter sterilized (0.2 μ m filter, Gelman Sciences). Ten larvae were used for each treatment. Controls were injected with 20 μ l sterile YS broth. In order to check for the promotion of nematode reproduction of *H. bacteriophora* by *P. luminescens* and *X. nematophilus* in the insect, living cells of H06, primary and secondary HNA, LN2, and A24 were injected at nematode inoculation.

RESULTS

From *in vitro* cultures in LCM bacteria-free DJs were harvested. The surface sterilization was successful, as controls never showed any growth of microorganisms. These DJs successfully invaded *G. mellonella* larvae from axenic cultures. However, the axenic *H. bacteriophora* had no pathogenic effect. Although 4% of the insects ($n = 50$) were dead after 3 days and 8% after 25 days, mortality probably was a result of cannibalism. When first checked after 3 days, dead individuals were injured and *G. mellonella* feeding on cadavers were observed. After 25 days, 50% of the insects had reached the adult stage. In the untreated control 48% reached adulthood. No mortality was recorded in controls without nematodes. All insects were dead when treated with DJs from monoxenic cultures with the specific symbiont. When treated with axenic DJs of *S. carpocapsae* all *G. mellonella* larvae had died after 3 days ($n = 30$). In contrast to insects infested with *H. bacteriophora*, the hemolymph of those treated with *S. carpocapsae* was liquefied.

H. bacteriophora found inside axenic *G. mellonella* had recovered after 3 days. Detected were 13.3 ± 6.4 nematodes per insect larva ($n = 10$). After 6 days some nematodes had developed to young hermaphrodites, others to hermaphrodites with eggs and first stage juveniles (J1) inside the uterus. Eggs and J1 were also found in the hemolymph. Development beyond the J1 stage was never observed. After 26 days, some hermaphrodites had survived in the insect pupae. Only dead hermaphrodites were found in the living insect adults. Injection of bacterial supernatant into *Galleria* larvae at nematode inoculation or 5 days after nematode inoculation did not induce further development of the J1; neither did the injection of YS broth. All axenic *Galleria* injected with H06 bacteria (20 μ l) at nematode inoculation were dead. H06 and HNA primary cells supported nematode reproduction; however, HNA secondary, LN2, and A24 cells did not.

Three days after infestation with *S. carpocapsae*, 9.4 ± 4.3 nematodes were found in the insects ($n = 10$). Invading juveniles recovered, developed to adults, and produced offspring. After 3 weeks no DJ had emigrated

from the cadaver. Dissection of the cadavers resulted in 5275.0 ± 6564.1 nematodes per insect ($n = 10$). The percentage of DJs was 6.7, of other larval stages 39.2, of males 11.9, and of females 42.2%. Even after 5 weeks, the percentage of dauer juveniles was only 9.9%.

DISCUSSION

It has been reported that some *Heterorhabditis* strains cultured with nonspecific *P. luminescens* isolates are unable to retain their symbionts in the gut. The resulting DJs were unable to kill *G. mellonella*, suggesting that *Heterorhabditis* lack the potential to synthesize entomotoxin (Han *et al.*, 1991; Gerritsen and Smits, 1993). The present results show that axenic *H. bacteriophora* lack the potential to produce metabolites which are toxic for axenic nonimmunized *G. mellonella* larvae. Even with more than 8 DJs inside an insect larva, the development to adults was not hampered. There were no signs of defense mechanisms against hermaphrodites. They may therefore just have died after offspring production, reaching their determined age. Under monoxenic conditions the hermaphrodites would have died earlier due to juveniles inside of the hermaphrodites feeding on its tissues (*endotokia matricida*). *S. carpocapsae* killed all larvae which indicate that entomotoxins are likely to be involved. The potential of axenic steinernematid DJs to kill insects has previously been reported for *S. carpocapsae* (Boemare *et al.*, 1982; Burman, 1982; Ehlers *et al.*, 1990; Simoes *et al.*, 1994) and *S. feltiae* (Ehlers *et al.*, 1997).

DJs of both species recovered and developed into feeding J3 stages as a response to food signals. The presented results further provide evidence that these food signals are present in the hemolymph of *Galleria* larvae, as reported by Strauch and Ehlers (1998), which used septic *G. mellonella*. Surprisingly, *S. carpocapsae* could continue its development and produced offspring DJs. This contradicts the results reported by Poinar and Thomas (1966). To produce axenic nematodes, they treated adults with antibiotics and then with merthiolate. Particularly the latter treatment might have sterilized the offspring of the adults, which might have been the reason for the unsuccessful reproduction. Another explanation is a possible lack of essential growth factors. In order to obtain axenic DJs, adults had been transferred into an autoclaved medium after the chemical sterilization. Due to the lack of growth factors this medium is unable to support development of axenic steinernematid nematodes. The axenic DJs obtained had probably developed on the content of the female's body (*endotokia matricida*). This might have produced offspring with little fat and other resources, which are sufficient only for development to adults but not for production of offspring. The bacteria-free DJs used in this study were reared on nonspecific *X. poinarii*. This method produced DJs with enough fat reserves, sufficient for the production of offspring which

were able to continue the development to DJs. In contrast to yields obtained under monoxenic conditions which reached a mean of 135.173 DJs (Poinar and Thomas, 1966), yields in axenic *G. mellonella* obtained during this investigation were low and the percentage of DJs did not surpass 10%. High nematode densities and low food signal concentrations in the presence of a nematode pheromone DRIF (Dauer Recovery Inhibiting Factor) induces the DJ formation in *Caenorhabditis elegans* (Golden and Riddle, 1984). Similar conditions were proposed for entomopathogenic nematodes (Popiel *et al.*, 1989; Fodor *et al.*, 1990; Strauch *et al.*, 1994; Han, 1996). The yields under axenic conditions were 5.275 nematodes after 3 weeks. Whether the lack of essential nutrients and/or the low concentration of DRIF has caused the low dauer formation and low yields needs further investigation.

Although the dauer juveniles of *H. bacteriophora* H06 developed into hermaphrodites with eggs and J1 in the hemolymph of the living insects, the J1 did not develop further. Injection of bacterial supernatants into the insect before and after infection with nematodes did not provide the essential nutrients necessary for a development beyond the J1 stage. Only the specific symbiont cells, and some nonspecific *P. luminescens*, contain the essential compounds necessary for the development of DJ offspring. These results again show that the symbiosis between heterorhabditid nematodes and *P. luminescens* seems to be more pronounced and intimate than the relation between steinernematids and their symbionts *Xenorhabdus* spp. in regard to pathogenicity and nutritive relation.

ACKNOWLEDGMENTS

The fellowship to Richou Han, Guangdong Entomological Institute, Guangzhou, China, by the Alexander von Humboldt Stiftung and the financial support by the European Commission (Pronema Project FAIR 3116) are gratefully acknowledged.

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